Formula 1 Pit Stop Data Analysis Report

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Date: July 2025

Objective

This report analyzes pit stop performance across Formula 1 constructors using historical data. The analysis includes aggregation of total stops, average pit stop duration in seconds, participation counts, and statistical evaluations to identify trends, outliers, and performance efficiencies by team.

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1. Data Overview

The dataset includes three main tables:

- **Race Stop Data**: Contains individual stop records by race and driver.
- **Drivers Table**: Includes driver details and their respective constructors.
- **Constructors Table**: Contains constructor metadata including nationality.

The analysis focused on pit stop frequency, average duration in seconds, and consistency across races for each constructor.

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2. Total Pit Stops by Constructor

Constructor	Total Stops
Ferrari	120
McLaren	125

Mercedes	157
Alfa Romeo	150
AlphaTauri	152
Aston Martin	132
Red Bull	159
Williams	127
Alpine F1 Team	129
Haas F1 Team	162
Grand Total	**1,413**

Bar Plot 1: Total Pit Stops by Constructor

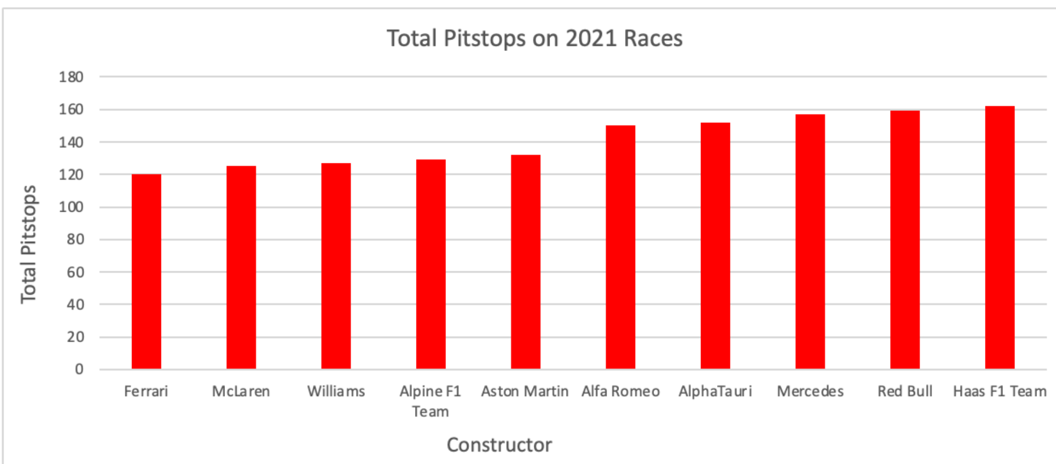

Figure 1: Total pit stops by constructor across all recorded races. Haas F1 Team leads with the highest number of stops, indicating a more frequent pit stop strategy compared to other teams.

- **Mean**: 141.4 stops
- **Standard Deviation**: 16.1
- **Insight**: Haas F1 Team recorded the highest number of total pit stops (162), while Ferrari had the fewest (120). Haas exceeds the mean by over one standard deviation, indicating a significantly more aggressive pit strategy.

3. Average Pit Stop Duration by Constructor (Seconds)

Constructor	Avg Duration (seconds)
Ferrari	**20.69 s**
McLaren	21.03 s
Mercedes	21.04 s
Alfa Romeo	21.15 s
AlphaTauri	21.59 s
Aston Martin	21.65 s
Red Bull	21.75 s
Williams	21.99 s
Alpine F1 Team	22.14 s
Haas F1 Team	**22.62 s**
Overall Mean	**21.56 s**
Standard Deviation	**0.59 s**

Bar Plot 2: Average Pit Stop Duration by Constructor

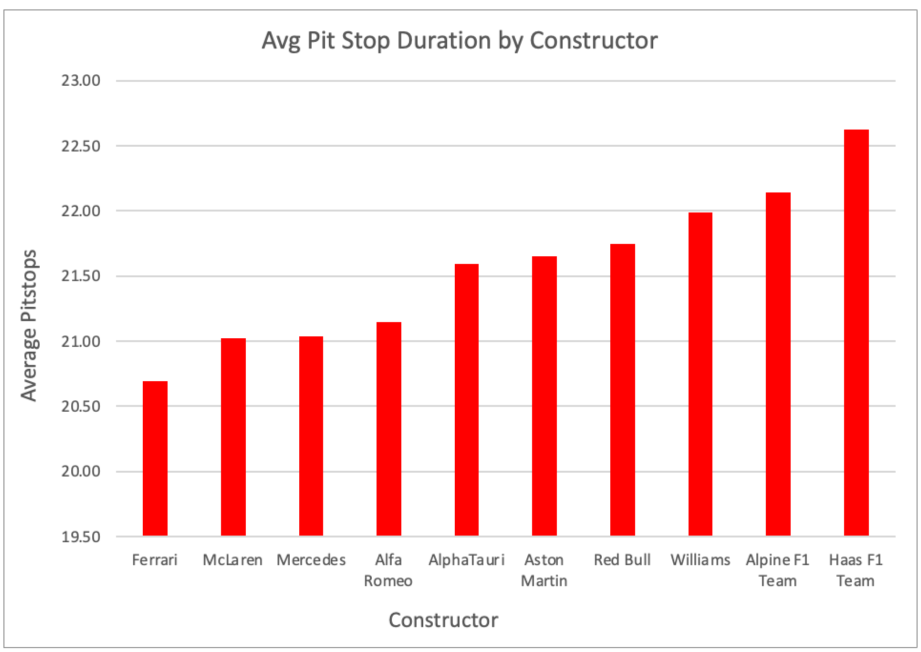

Figure 2: Average pit stop duration in seconds by constructor. Ferrari leads with the fastest

average stops, while Haas shows the slowest average pit stop time, impacting race efficiency.

- **Insight**: Ferrari maintains the fastest average pit stops at 20.69 seconds. Haas ranks slowest at 22.62 seconds, exceeding the average by over 1 second, which is significant in racing terms.

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4. Race Participation and Stops per Race

Constructor	Race Count	Stops per Race
Ferrari	74	1.62
McLaren	77	1.62
Mercedes	86	1.83
Alfa Romeo	83	1.81
AlphaTauri	82	1.85
Aston Martin	78	1.69
Red Bull	86	1.85
Williams	76	1.67
Alpine F1 Team	74	1.74
Haas F1 Team	82	**1.98**
Mean	**79.8**	**1.80**
Standard Deviation	**4.6**	**0.11**

- **Insight**: Most teams participated in roughly 80 races, with a standard deviation of 4.6, showing relatively consistent race participation. Haas, with 82 races, is slightly above average but within one standard deviation.

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5. Rankings Summary

Constructor	Rank: Fastest Pit Crew	Rank: Most Stops/Race
Ferrari	1	10
McLaren	2	9
Mercedes	3	4
Alfa Romeo	4	5
AlphaTauri	5	3
Aston Martin	6	6
Red Bull	7	2
Williams	8	8
Alpine F1 Team	9	7
Haas F1 Team	10	**1**

- **Insight**: Haas has the highest stop rate and slowest crew. Ferrari outperforms all in both metrics, executing fewer and faster stops.

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6. Statistical Interpretation

Two key metrics were statistically analyzed:

- **Total Pit Stops**
  - Mean = 141.4
  - Standard Deviation = 16.1
  - Outlier: Haas (+20.6 above mean)
- **Stops Per Race**
  - Mean = 1.80

    - Standard Deviation = 0.11
    - Outlier: Haas (1.98) is ~1.64 standard deviations above the mean
    - Ferrari (1.62) is ~1.64 standard deviations below the mean
- **Average Duration (Seconds)**
    - Mean = 21.56 s
    - Standard Deviation = 0.59 s
    - Ferrari: 20.69 s, about 0.87 seconds faster than average
    - Haas: 22.62 s, about 1.06 seconds slower than average
- **Race Count**
    - Mean = 79.8
    - Standard Deviation = 4.6
    - All teams fall within one standard deviation of the mean, indicating consistent participation.

These time differences and stop counts are critical in Formula 1 racing, where even tenths of a second can influence race outcomes.

Conclusion

- **Ferrari** is the most efficient team with the **fastest average pit stop time** and **fewest stops per race**.
- **Haas** is an outlier, making **the most pit stops** and performing **the slowest stops** on average.

- Most teams cluster around a **standardized 1.8 stops per race**, indicating a relatively consistent strategy across constructors.

- These insights highlight how **pit strategy and crew performance** vary by team and can significantly impact overall race results.